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Car Body Design - Projects

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Report 1

Wheel arch and ground clearance sizing

1.1 Introduction

The aim of this report is to establish the dimensions of the front and rear wheel arches (starting from a top view analysis) and the ground clearance sizing on loading conditions (starting from a lateral view analysis) for a commercial vehicle.

1.2 Method: detailed description of the design procedures

Among the commercial vehicles, the Volkswagen Golf 4 (2003) has been chosen from catalogs. Relevant parameters of the selected vehicle are the following ones:

Golf 4 (2003)	**Parameter**	**Value**	**Unit**
Parameters	Wheelbase (l)	2518	mm
	Track (t)	1539	mm
	Size of tires	225/40 R18	/
	R_1	4680,5	mm
	Number of occupants	5	/
	Mass (empty)	1433	kg
	Mass (full)	1474	kg
Weight distribution (empty load)	Front	63,50	%
	Rear	36,50	%

Table 1.1: Catalog parameters

There are several wheel sizes to choose from, the R18 has been selected to take into account the largest possible bulk.

Some images of the vehicle are reported.

Figure 1.1: Volkswagen Golf 4

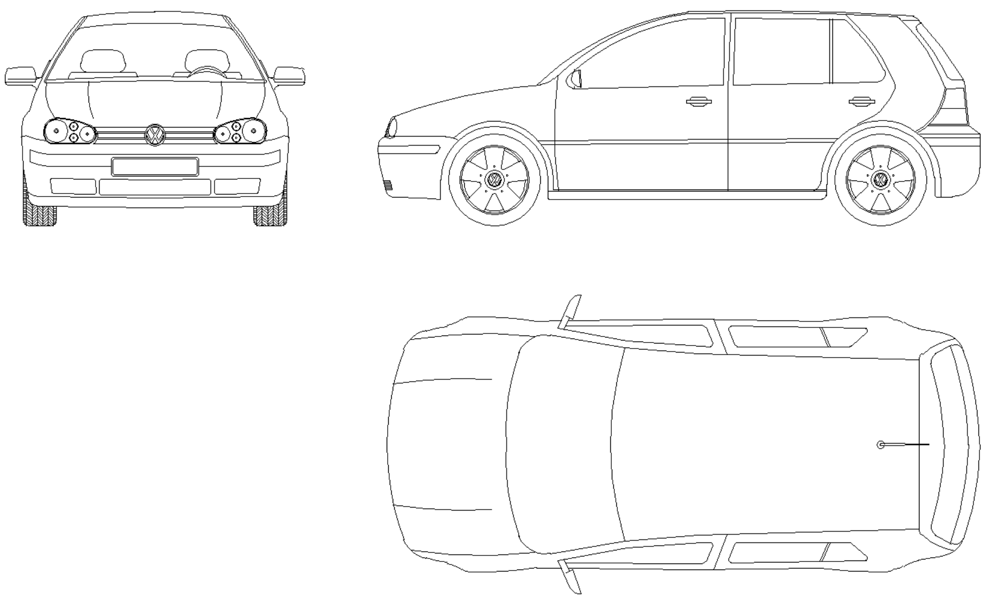


Figure 1.2: Cad

1.2.1 Sizing of the front wheel arch

From a top view analysis, illustrated in figure 1.3, it is possible to compute the kinematic steering dimensions (the outer angle δ_1, the inner angle δ_2 and the steering radius R) using the following relations:

$$l = \left(R_1 - \frac{t}{2}\right)\tan\delta_1 \implies \delta_1 = \arctan\left(\frac{l}{R_1 - \frac{t}{2}}\right) \tag{1.1}$$

$$l = \left(R_1 + \frac{t}{2}\right)\tan\delta_2 \implies \delta_2 = \arctan\left(\frac{l}{R_1 + \frac{t}{2}}\right) \tag{1.2}$$

$$R = \sqrt{\left(R_1 + \frac{t}{2}\right)^2 + l^2} \tag{1.3}$$

Considering the values of l, R_1, and t (taken from the table 1.1) the following results are obtained from the previous equations:

$$\begin{cases} \delta_1 = 32,77° \\ \delta_2 = 24,80° \\ R = 6004\,mm \end{cases}$$

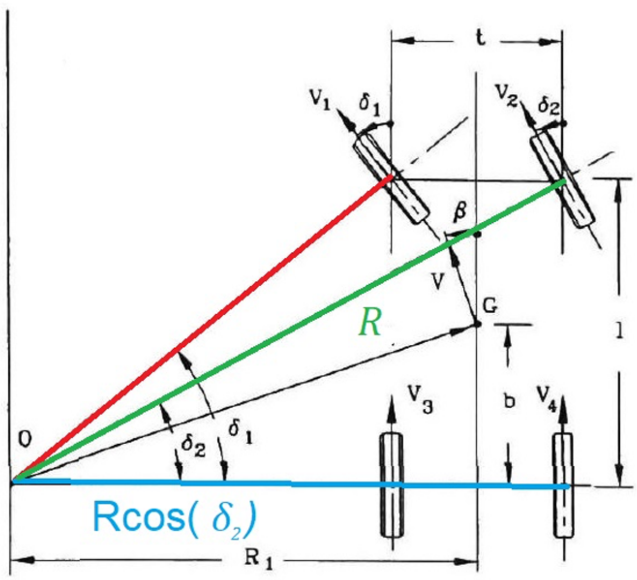


Figure 1.3: Kinetic steering

Once the steering angles and the steering radius are known, the size of the tire from table 1.1 can be used to estimate the characteristic dimensions of the front wheel arch in top view.

Considering the parameter 225/40 R18, the largest size of the wheel, with added the space for snowchains and dirt layer, is computed:

$$Z = \underbrace{\frac{18}{2} \cdot 25,4 + \frac{40}{100} \cdot 225}_{\text{without snowchains: } 318,6mm} +25 = 343,6\ mm$$

1.2.2 Sizing of the rear wheel arch

Since the rear tires do not steer (figure 1.3), it is necessary to calculate only the gap between wheel and arch.

1.2.3 Sizing of the ground clearance

The evaluation of the possible load conditions of the vehicle is needed in order to define properly the ground clearance displacements. This aspect will take into account variable different loads configuration: each case takes into account the addition of passengers and luggages, whose mass values in kg are defined in figure 1.2.

Load type	**Mass [kg]**
Passenger	70
Luggage	10

Table 1.2: Load assumption for passenger and luggage

Considering that the vehicle is homologate for 5 passengers, there will be 6 load conditions: first load case doesn't take into account any passengers while other cases are defined with a progressively increasing number of occupants from 1 to 5, with each occupant accounted for with additional luggage.

Load conditions	$m_f[kg]$	$m_r[kg]$	$m_l[kg]$
A	0	0	0
B	70	0	10
C	140	0	20
D	140	70	30
E	140	140	40
F	140	210	50

Table 1.3: Load conditions

In order to evaluate the load distribution between the axles, it is essential to estimate the application points of each load with respect to the vehicle's center of gravity and the luggage's center of mass. Figure 1.4 shows the necessary dimensions for the calculations.

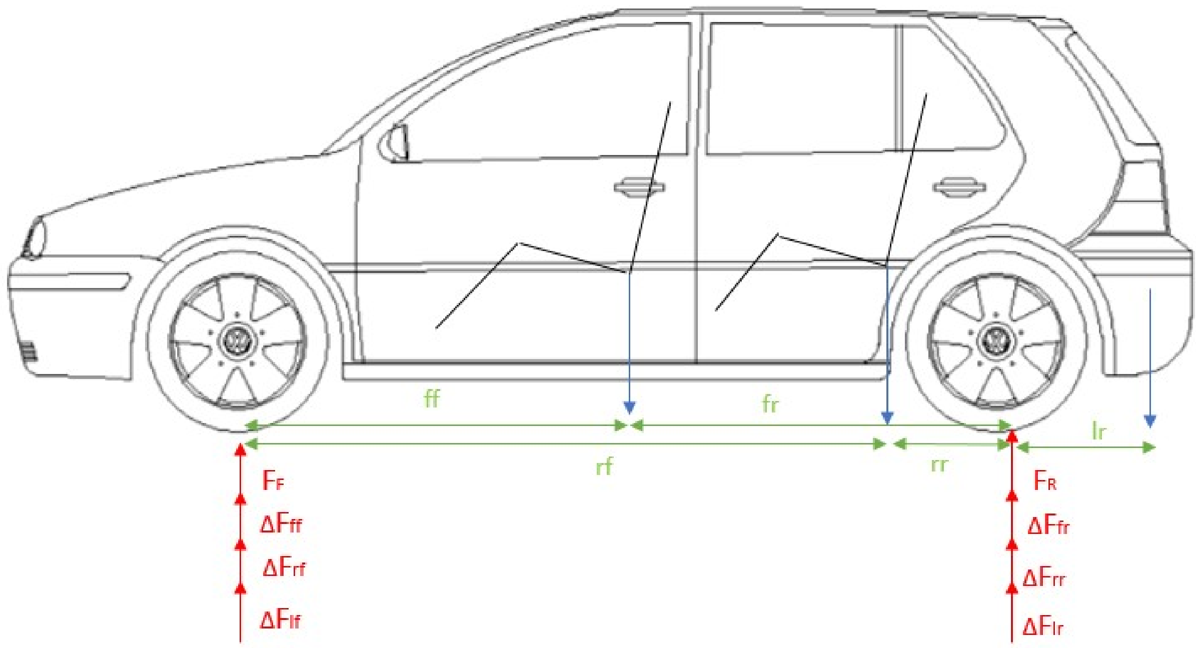


Figure 1.4: Scheme of load distribution

In table 1.4 are reported the values of the distances. The first letter refers to the row and the second one is related to the axle (for example, fr is the distance between front row and rear axle).

Indicator	**Value** [mm]
ff	1262
fr	1256
rf	2200
rr	318
lf	2951
lr	433

Table 1.4: Vehicle dimensions

The following equations are used to compute the total force applied on the front and the rear axles, in each load condition:

$$\Delta F_{ff,i} = 9,81 \cdot m_{f,i} \cdot \frac{\overline{fr}}{W_b} \tag{1.4}$$

$$\Delta F_{rf,i} = 9,81 \cdot m_{r,i} \cdot \frac{\overline{rr}}{W_b} \tag{1.5}$$

$$\Delta F_{lf,i} = 9,81 \cdot m_{l,i} \cdot \left(1 - \frac{\overline{lf}}{\overline{lf} + \overline{lr}}\right) \tag{1.6}$$

$$\Delta F_{fr,i} = 9,81 \cdot m_{f,i} \cdot \frac{\overline{ff}}{W_b} \tag{1.7}$$

$$\Delta F_{rr,i} = 9,81 \cdot m_{r,i} \cdot \frac{\overline{rf}}{W_b} \tag{1.8}$$

$$\Delta F_{lr,i} = 9,81 \cdot m_{l,i} \cdot \frac{\overline{lf}}{\overline{lf} + \overline{lr}} \tag{1.9}$$

Summing all the forces, the resultants on the front and the rear axle are determined for all the load cases (table 1.5).

Loadcase	$\boldsymbol{\Delta F_{f,i}[N]}$	$\boldsymbol{\Delta F_{r,i}[N]}$
A	0	0
B	355	430
C	710	859
D	809	1545
E	908	2230
F	1008	2916

Table 1.5: Force applied on the axles

Once the load conditions are defined, the aim is to compute the bending stiffness of each axle and on this purpose the quarter vehicle model of the suspension must be analyzed (figure 1.5).

The contribute of the unsprung masses to the quarter model can be neglected: this assumption simplifies the system, allowing to consider only the mass of the axle and the suspension (modeled as damper and spring).

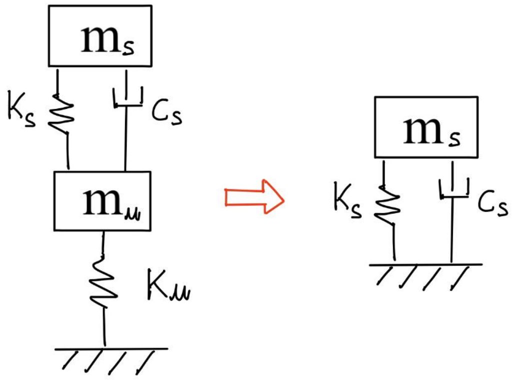


Figure 1.5: Quarter car model

At this point, the axles suspensions stiffness can be computed starting from the following equation:

$$f = \frac{1}{2\pi} \cdot \sqrt{\frac{K_s}{M_s}} \tag{1.10}$$

where f is the frequency, M the mass and K the bending stiffness.

In order to do the calculations, the total mass distribution on the two axles and the operating natural frequencies are required.

For both front and rear axles, ΔF related to the worst load case (F) has been divided by the gravity acceleration g, becoming a mass contribution. The sum of this contribution with the mass on each axle at empty load is finally computed. The results are reported in table 1.6.

	Axle	**Mass [$\boldsymbol{kg}$]**
M_{sf}	front	1038.90
M_{sr}	rear	835.35

Table 1.6: Weight distribution (case F)

The missing data for the evaluation of the bending stiffness from equation 1.10 are the operating natural frequencies: their value is obtained through some considerations about the discomfort diagram, correlating vertical acceleration and frequency (figure 1.6).

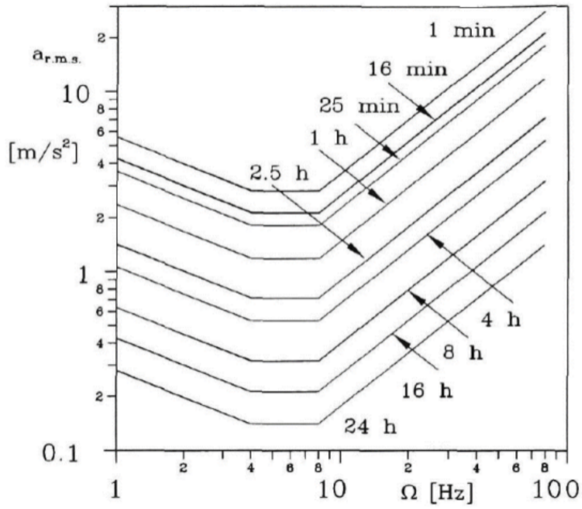


Figure 1.6: Discomfort diagram

It's better to fix the operating frequency in the portion of the graph where the following criteria is met: if an increase of the frequency is provided, in order to maintain the same level of discomfort, the acceleration must be reduced.

This criteria is pleased in the right portion of the graph, for frequencies between $1Hz$ and $4Hz$, while other portions provide negative effects such as motion sickness (in case of negative frequency) or high sensitivity to vibration (which implies high discomfort for the passengers).

That is the reason why $1.5Hz$ is chosen for front suspension in our calculations. For the rear suspension, $1.65Hz$ is considered in order to improve the displacement which could affect the rear axle in condition of maximum load.

Once axle masses and operating natural frequencies are fixed, the evaluation of bending stiffness is done through the following equations:

$$K_{sf} = M_{sf} \cdot (2\pi \cdot 1.5)^2 \tag{1.11}$$

$$K_{sr} = M_{sr} \cdot (2\pi \cdot 1.65)^2 \tag{1.12}$$

The values are reported in table 1.7.

	Axle	**stiffness [N/mm]**
K_{sf}	front	92.28
K_{sr}	rear	89.78

Table 1.7: Bending stiffness

Therefore, it's possible to evaluate the displacements D and the ground clearance GC on the front and on the rear, using the following equations.

$$D_{f,i} = \frac{\Delta F_{f,i}}{K_{sf}} \tag{1.13}$$

$$D_{r,i} = \frac{\Delta F_{f,i}}{K_{sr}} \tag{1.14}$$

The results are shown, respectively, in tables 1.8 and 1.9. Whereas, the following equations are used for the ground clearance computations:

$$GC_{f,i} = 102 - D_{f,i} \tag{1.15}$$

$$GC_{r,i} = 102 - D_{r,i} \tag{1.16}$$

where 102 (mm) represents the initial ground clearance (from Volkswagen catalog) at empty load. The results are shown in the table 1.9.

$[mm]$	$\boldsymbol{D_{f,i}}$	$\boldsymbol{D_{r,i}}$
A	0.00	0.00
B	3.85	4.79
C	7.70	9.57
D	8.77	17.21
E	9.85	24.84
F	10.92	32.48

Table 1.8: Table of displacement

$[mm]$	$\boldsymbol{GC_{f,i}}$	$\boldsymbol{GC_{r,i}}$
A	102.0	102.0
B	98.15	97.21
C	94.3	92.43
D	92.23	84.79
E	92.15	77.16
F	91.08	69.52

Table 1.9: Table of ground clearance.

1.3 Results and discussion

1.3.1 Front wheel arch

As it is possible to see from figure 1.7, the sizing of front wheel arch comes to conclusions considering the expected gap of 25 mm for snow-chains and dirt layer, the displacement of the ground clearance previously computed on the front axle (10.92 mm) and an additional travel of 35 mm for the suspension.

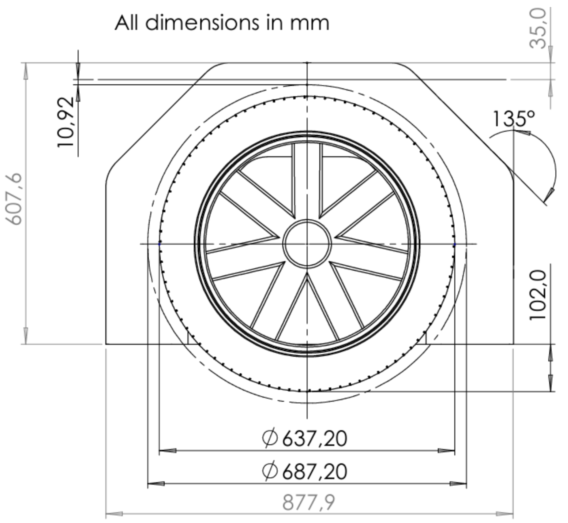


Figure 1.7: Side view of the front wheel arch

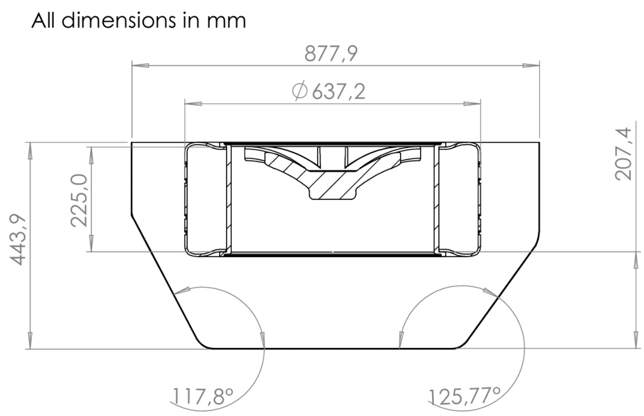


Figure 1.8: Bottom view of the front wheel arch

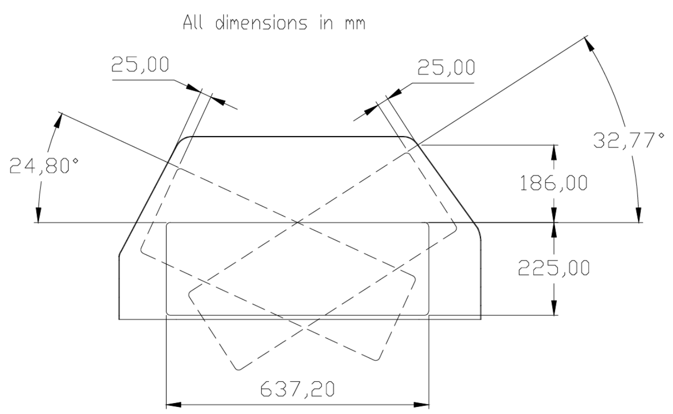


Figure 1.9: Maximum steering angles configuration in bottom view

Figure 1.10: Rendering of the front wheel arch where δ_1 is considered

1.3.2 Rear wheel arch

As for the front wheel arch, $25\,mm$ clearance is considered to take into account snow-chains e dirt accumulation. A gap on the rear, related to the ground clearance displacement in the worst load case, has been previously computed (32.5 mm for the rear) and an additional $35\,mm$ travel for the suspension is also taken into account.

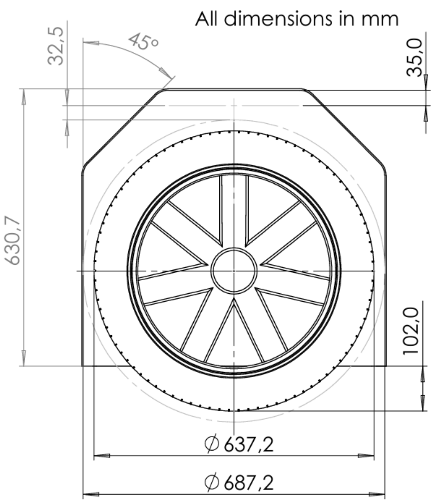


Figure 1.11: Side view of the rear wheel arch

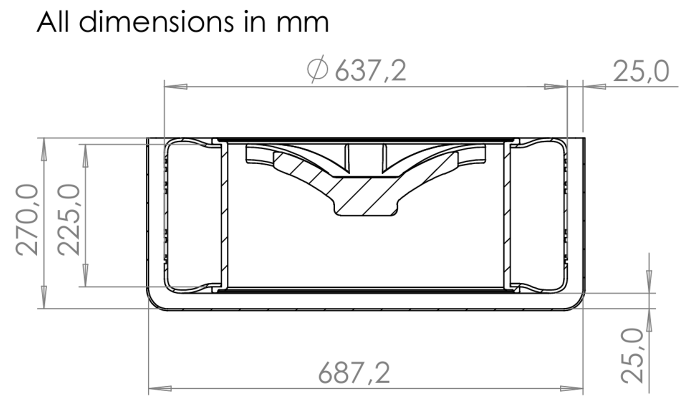


Figure 1.12: Bottom view of the rear wheel arch

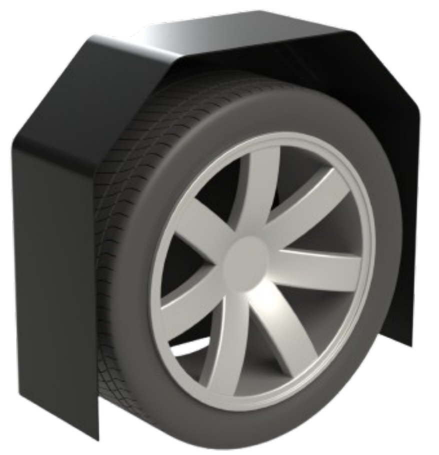


Figure 1.13: Rendering of the rear wheel arch where δ_1 is considered

1.3.3 Ground clearance

Considering displacements computed in paragraph 1.2.3 a draw illustrating the variations of ground clearance at different load conditions.

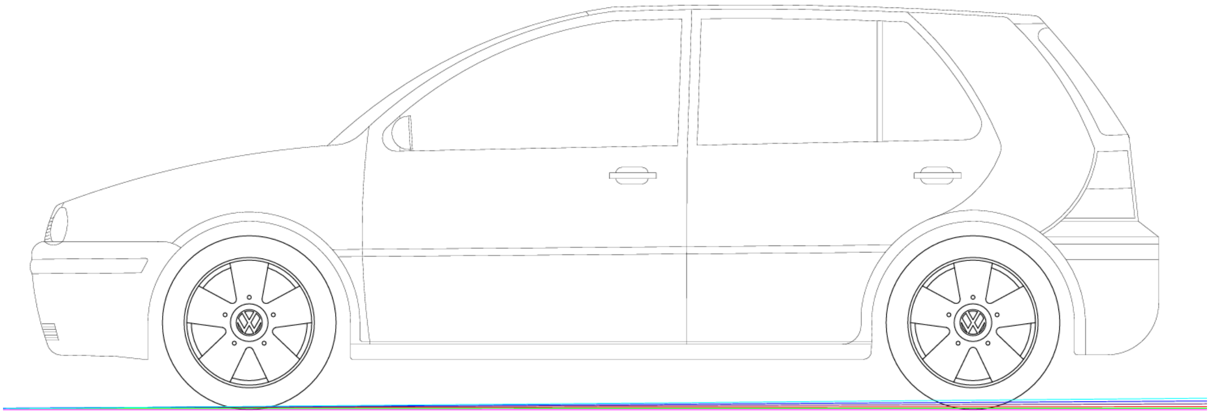


Figure 1.14: Car displacement at different load condition

Zooming on tyre contact area is evident that rear axle has an higher displacement than the front, due to the fact that the majority of the the loaded weight (passengers and luggage) is located behind the center of gravity. In addition, focusing on front axle, case A, B and C show a bigger increment in front displacement, because the load is mostly applied ahead the center of gravity, instead of the other cases.

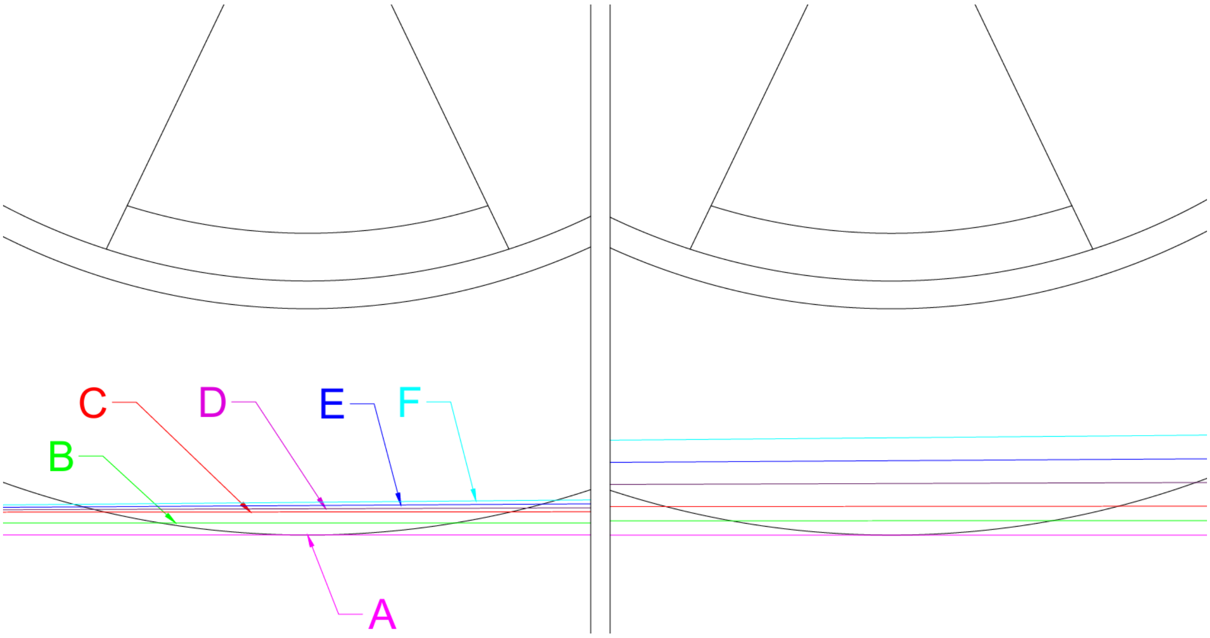


Figure 1.15: Focus on front (left picture) and rear (right picture) axle

1.4 Conclusion

This project successfully demonstrates a thorough and systematic approach to determine the optimal dimensions for the front and rear wheel arches, as well as the ground clearance, of the Volkswagen Golf 4 (2003). The design process accounted for a wide range of critical factors, including steering angles, load distributions, and suspension travel requirements, ensuring the vehicle performs efficiently and safely in diverse operational conditions.

The result is shown in the following figure.

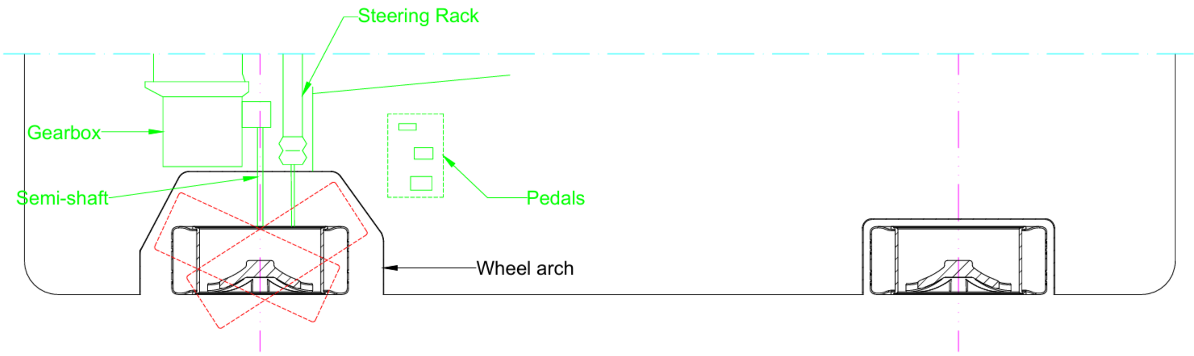


Figure 1.16: Upper view of front and rear wheel arch

Special considerations were given to real-world scenarios, such as accommodating snow chains, dirt accumulation, and varying load cases, from an empty vehicle to full capacity with passengers and luggage. The methodology also included calculations for axle stiffness, displacement, and ground clearance adjustments under maximum load conditions, ensuring durability and reliability of the vehicle's structure over time.

The final design strikes an effective balance between practicality, safety, and performance, meeting industry standards and enhancing the vehicle's adaptability to challenging road conditions.

This work highlights the importance of integrating engineering principles with real-world requirements to achieve a functional, efficient, and user-focused automotive design.

Report 2

Battery dimensioning for electric vehicles

2.1 Introduction

The aim of the report is to design a battery system for commercial electric vehicles: starting from a sizing process in terms of nominal voltage and energy, the layout of cells and modules is going to be defined in order to obtain fundamental results. These results are mainly centered on the following aspects:

- thermal management system;
- power and signal electrical connections;
- gravimetric and volumetric energy density analysis at cell and pack level.

2.2 Method

The battery studied is conceived to fit in high-performance cars, in order to guarantee the following nominal values:

Rated Voltage	800	V
Battery energy	96	kWh
Battery capacity	120	Ah

Table 2.1: Battery pack data

Where the battery capacity in table 2.1 is computed as it follows:

$$battery\ capacity = \frac{battery\ energy}{rated\ voltage}$$

2.2.1 Battery layout

A battery layout **cell-to-module** has been chosen. The decision to adopt a cell-to-module (CTM) architecture for the battery pack is driven by several key factors.

First benefit consists in simplified design and increased reliability of the CTM architecture: this choice improves modularity and also makes thermal and mechanical management simpler. Additional advantage concerns faulty modules, which don't compromise too much the functionality of the battery pack if one of the single module is damaged.

The preliminary design of a single module must guarantee the values reported in table 2.2.

Number of modules	4	/
Rated voltage	400	V
Module capacity	60	Ah

Table 2.2: Target of the module configuration

Four modules have been considered, as it is a good compromise for balancing and managing the battery pack. The module connection is computed as Xs and Yp from the following relations:

$$X = \frac{battery\ rated\ voltage}{module\ rated\ voltage} = \frac{800\,V}{400\,V} \qquad \Longrightarrow \qquad X = 2$$

$$Y = \frac{battery\ capacity}{module\ capacity} = \frac{120\,Ah}{60\,Ah} \qquad \Longrightarrow \qquad Y = 2$$

As consequence, $2s2p$ describes the series/parallel connection of the modules.

2.2.2 Type of cell

The driving features of a battery for high performance car must be high volume efficiency and easy thermal management.

Prismatic cells ensure high volume efficiency with the disadvantage of a difficult thermal management whereas cylindrical ones provide good cooling performances at the cost of low volumetric efficiency.

Among the common types of cells, **pouch type** has been selected: this choice ensures both volumetric and thermal efficiency. They are more expensive, but this choice is followed by the assumption that the cost is not a binding feature for a high-performance car. In addition, pouch cells guarantee a low mass and custom-built packaging, with soft polymer aluminum plastic film.

Feature	**Pouch**	**Prismatic**	**Cylindrical**
Capacity (Ah)	$0.5 \div 100+$	$20 \div 200$	$1 \div 5$
Gravimetric Energy Density (Wh/kg)	$200 \div 300$	$180 \div 250$	$150 \div 250$
Volumetric Energy Density (Wh/L)	$500 \div 600$	$350 \div 500$	$300 \div 400$
Mechanical strength	Low	High	Very High
Design Flexibility	Very High	Medium	Low
Unit Cost	High	Medium	Low
Ease of Production	Low	Medium	High

Table 2.3: Comparison of pouch, prismatic, and cylindrical cells.

For our application the *LG Chem E66A* (figure 2.1) has been chosen. It guarantees low cell mass and a very high nominal capacity, being a decent fit considering that a single pouch cell ($65Ah$) reaches the capacity target of the module ($60Ah$). Its parameters are shown in figure 2.4.

Figure 2.1: Pouch cell - LG Chem E66A

Parameter	**Value**	**Unit**
Dimensions (l x h x w)	350 x 104 x 11.7	mm
Effective volume	0.39	l
Cell mass	0.897	kg
Voltage (min ÷ max)	$2.5 \div 4.2$	V
Nominal voltage	3.7	V
Nominal capacity	65	Ah
Energy density	232.2	Wh
Gravimetric energy density	259	Wh/kg

Table 2.4: LG Chem E66A data

2.2.3 Module layout

In order to accomplish the fixed target for the module, the cells disposition for a single module is computed (once nominal voltage and nominal capacity of a single pouch are known).

$$X = \frac{module\ rated\ voltage}{cell\ nominal\ voltage} = \frac{400\,V}{3.7\,V} \qquad \Longrightarrow \qquad X = 108$$

$$Y = \frac{module\ capacity}{cell\ nominal\ capacity} = \frac{60\,Ah}{65\,Ah} \qquad \Longrightarrow \qquad Y \simeq 1$$

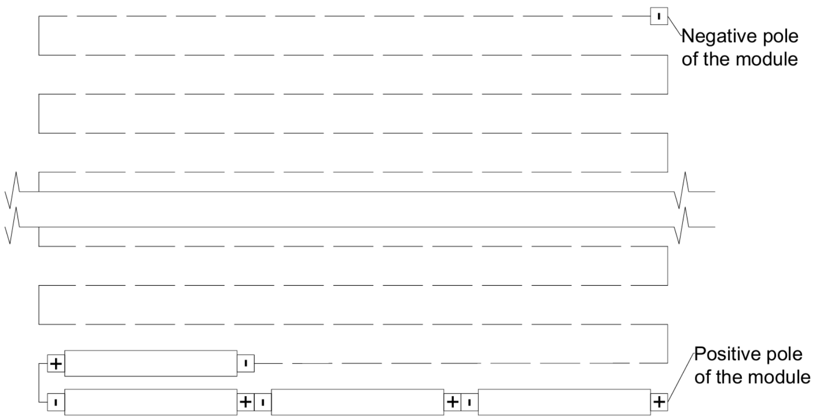
As consequence, $108s1p$ describes the series/parallel connection of the pouch cells in a single module.

2.3 Results and discussion

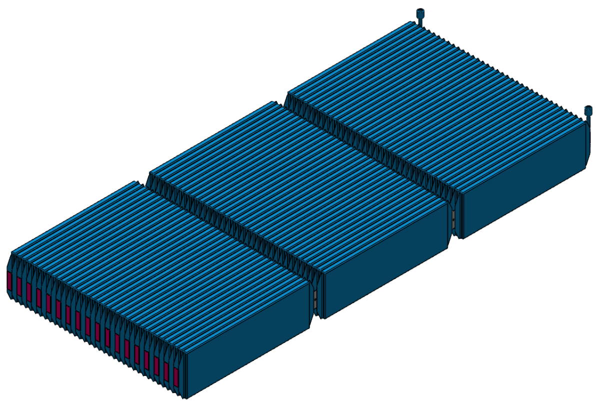
2.3.1 Module description

For a proper design of the modules, the main targets have been taken into account simultaneously, so the design has not followed a straight path but all decisions were constantly adjusted in order to reach the optimal result.

In figure 2.2a, the arrangement of the cells is schematized. The design aim is to simplify electrical connections while minimizing space and material usage. The pouch cells are directly connected to each other within the same row using spot welds, while the rows themselves are interconnected with intermediate copper connectors secured with spot welds, too. In figure 2.2b the result is shown.



(a)

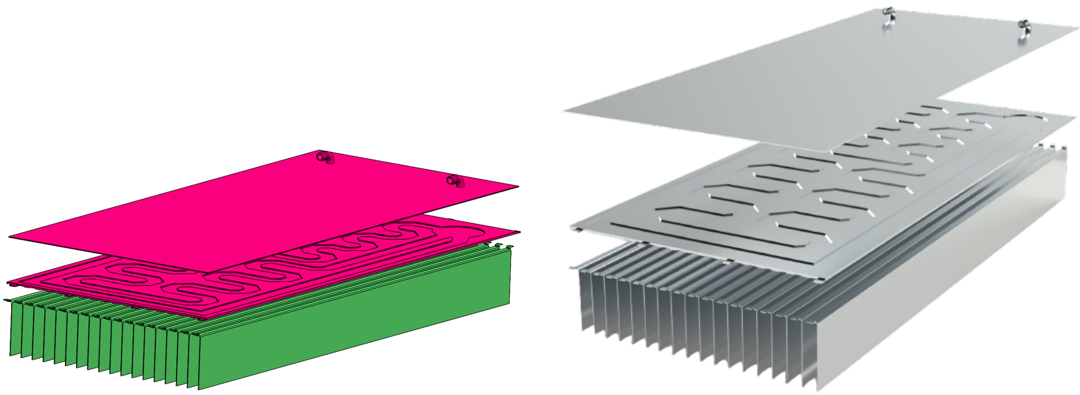


(b)

Figure 2.2: Battery arrangement in the module

The cooling plate is selected as thermal management system: its placement and the arrangement of the cells within the module have been designed. To achieve this, a single cooling plate is positioned across the entire module, with the pouches arranged vertically. This configuration allows us to determine the number of cells in terms of width, length, and height, which is set at 36x3x1. In order to justify this choice, the following assumptions have been considered:

- in height, the single layer ensures that every cell makes direct contact with the cooling plate, enhancing thermal efficiency;
- placing three cells in length results in a total module length of approximately 1 meter. Combined with an additional module (using the 2x2 configuration), it ensures that the total length does not exceed 2.5 meters;
- the choice of 36 cells in width is a necessary design constraint to meet the nominal capacity target.

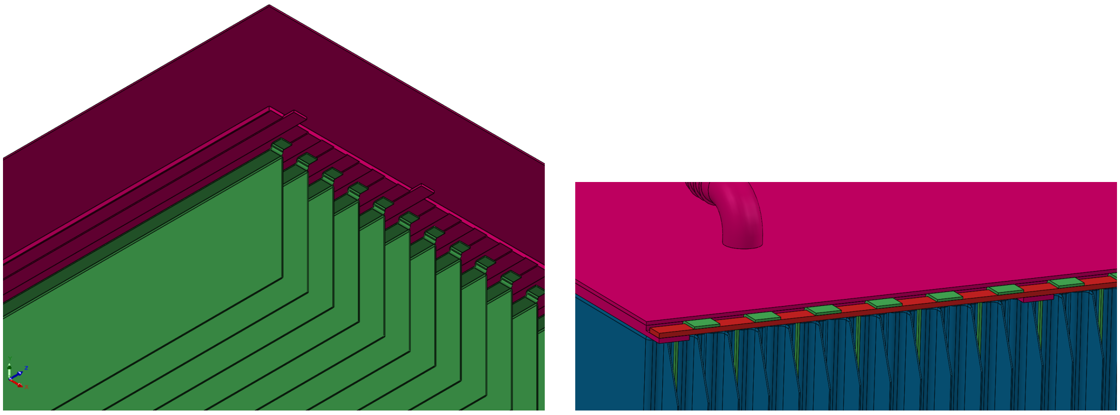


(a) CAD (b) Rendering

Figure 2.3: Cooling system assembly

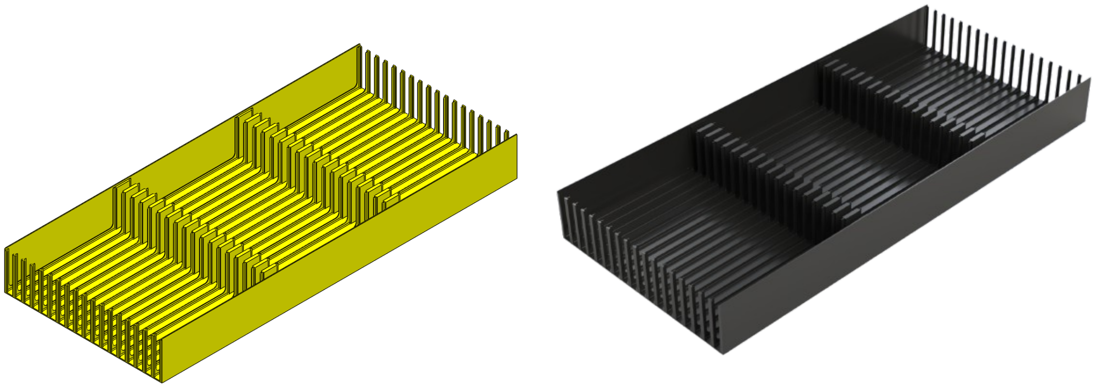
To enhance thermal efficiency, aluminum sheets ($1\,mm$ thick) are placed in between of every couple of cells, ensuring that each one of them has one side in contact with the sheets. The aluminum sheet is wrapped around the top of each cell, in order to get direct contact with the cooling plate (figure 2.4a), and fastened with a plastic bar (figure 2.4b) which is constrained between the lower part of the cold plate and the fin. The result is showed in figure 2.3.

The next step concerns the design of the optimal case to host cells and cooling plate. ABS (Acrylonitrile Butadiene Styrene) is the selected material, due to its excellent durability, high-temperature resistance and lightweight properties. Grooves are incorporated into the bottom plane to hold two cells in place securely, with a $1\,mm$ gap with respect to their lateral surfaces. This gap ensures enough space to allow thermal expansion during operating condition.



(a) (b)

Figure 2.4: Cooling system details

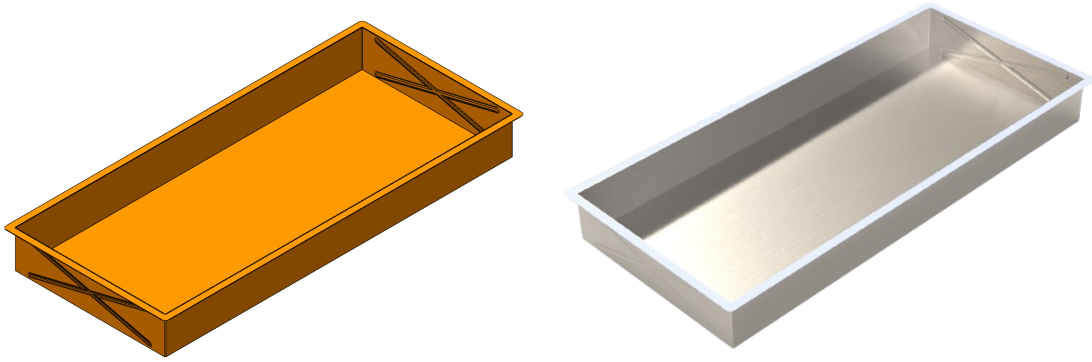


(a) CAD (b) Rendering

Figure 2.5: Plastic case for pouch housing

With the aim of protecting all the components from external environment and improving the reliability of the battery module, a case is developed: the expected outcome is a case as lighter as possible while able to sustain the heavy weight of battery. Moreover, from the structural point of view, during the preliminary phase of design, another goal concerns the development of a frame which is able to enhance the torsional stiffness of the chassis where the battery pack is bolted.

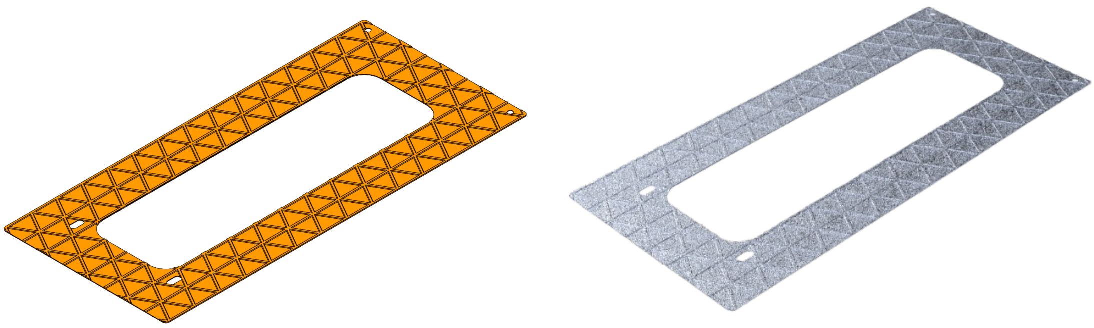
As shown in figure 2.6, the bottom part of the case is a box, realized from a steel sheet 1.5 mm thick. The choose of the material was driven by the necessity to support the huge weight of the component inside it (around 100 kg). Aluminum was taken into account but, after same consideration, it was discarded due to its worse mechanical characteristic. Furthermore, a deep drawing is realized in two faces of the box, to accomplish an increment of the stiffness.



(a) CAD (b) Rendering

Figure 2.6: Lower part of module case

In figure 2.7 the upper part of case module is shown. It is an aluminum plate realized through the cast process with a minimum thickness of 1,5 mm. Triangle ribs, also $1.5mm$ thick, were added to enhance the stiffness without increase too much the weight. The shape of the part was developed to implement the concept of the clamped beam: The upper part of the case, bolted with the frame of the battery back, assumes a structural role.



(a) CAD (b) Rendering

Figure 2.7: Upper part of module case

Focusing on module assembling, the figure 2.8 shows the exploded view and the figure 2.10 shows the final assembly of the module.

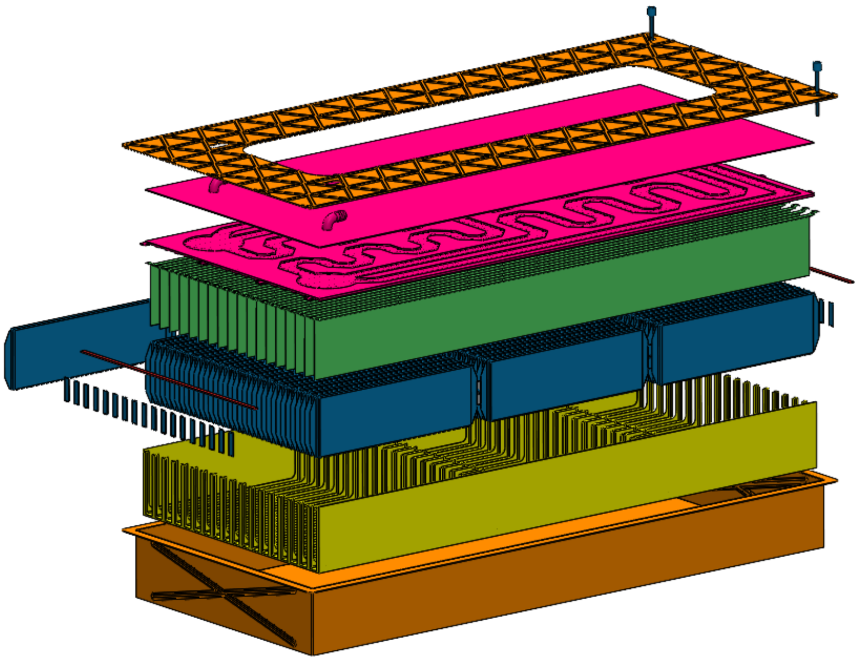
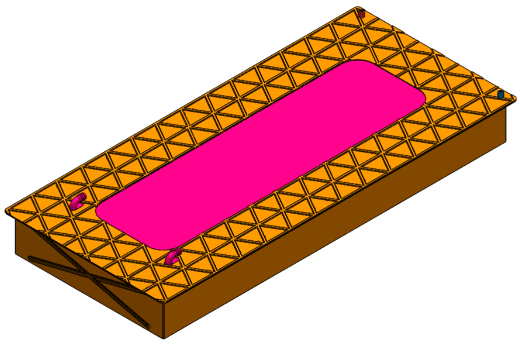
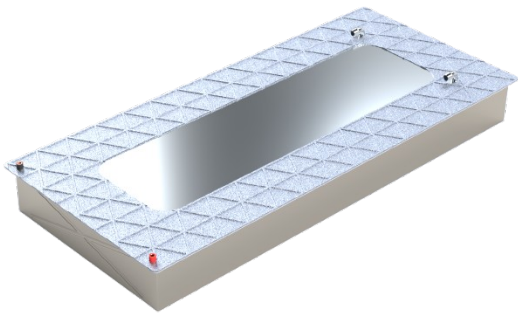


Figure 2.8: Exploded view of battery module assembly



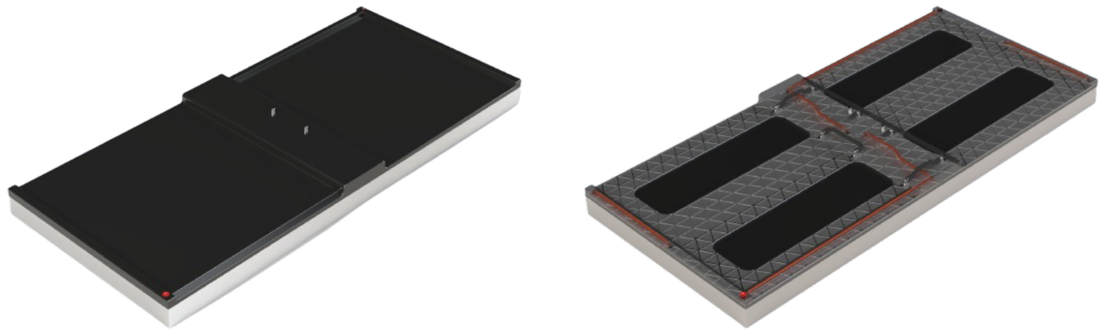
(a) CAD



(b) Rendering

Figure 2.9: Final assembly of battery module

2.3.2 Battery pack description



(a) (b)

Figure 2.10: Final assembly of battery module

Following the idea of making a battery pack able to fit between the longitudinal frames of a vehicle, the modules have been put as in figure 2.11.

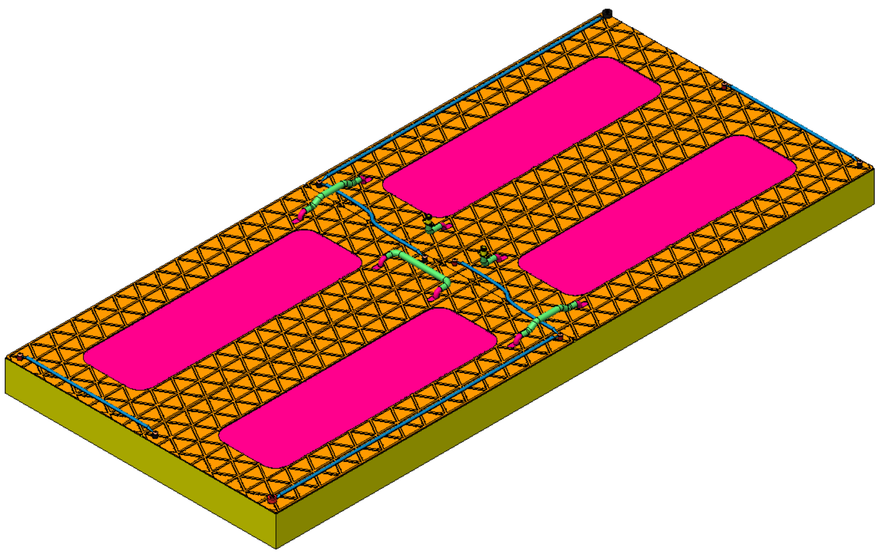


Figure 2.11: Power and cooling connections

This also made easier to electrically connect them, recalling that the chosen configuration is 2S 2P. As shown in the schematization reported in figure 2.12 and on blue connections in figure 2.11, the two series (green connections) are between the modules that share the longer lateral surfaces, while the two parallel (red and black connections) are between the modules that share the shorter surface.

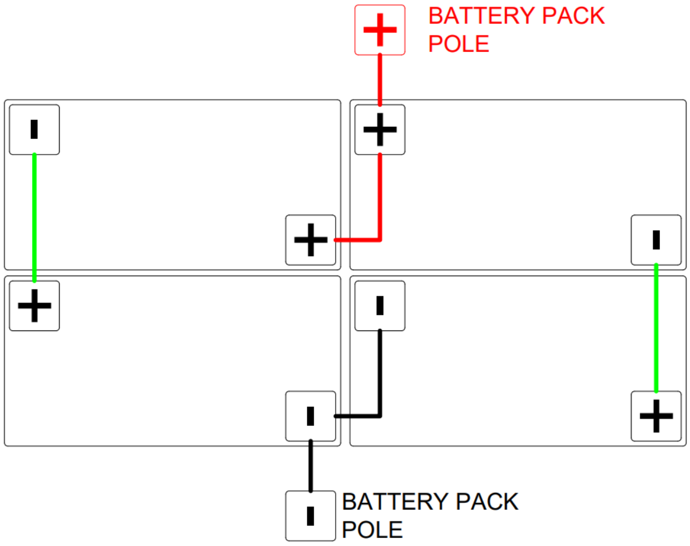


Figure 2.12: Schematization of battery pack

As reported in figure 2.11 (green tubes) the cooling system features input and output tubes for the coolant arranged in a circular pattern. The primary inlet and outlet are located at the center of the battery pack, with each module's input connected to the output of the adjacent module. Then the general inlet and outlet are inserted into the battery case and get out of the battery pack to get connected to the external part of the cooling and heating system.

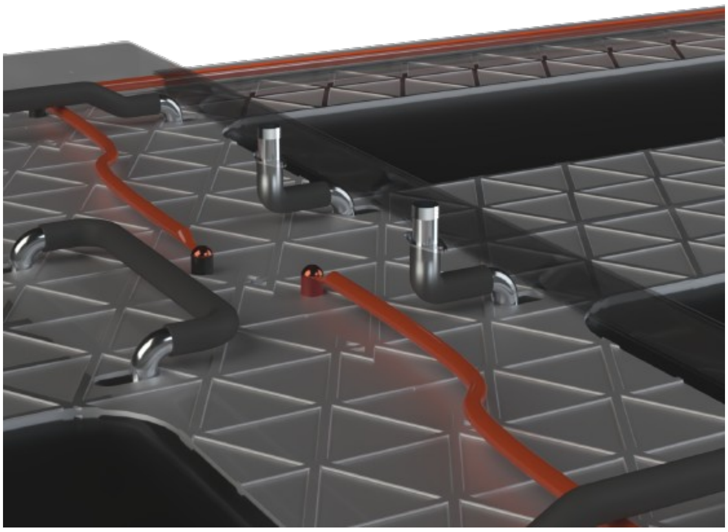


Figure 2.13: Details of battery pack

The battery case is made out of plastic, in particular ABS (as for the module case), to guarantee a certain stiffness and heating resistance.

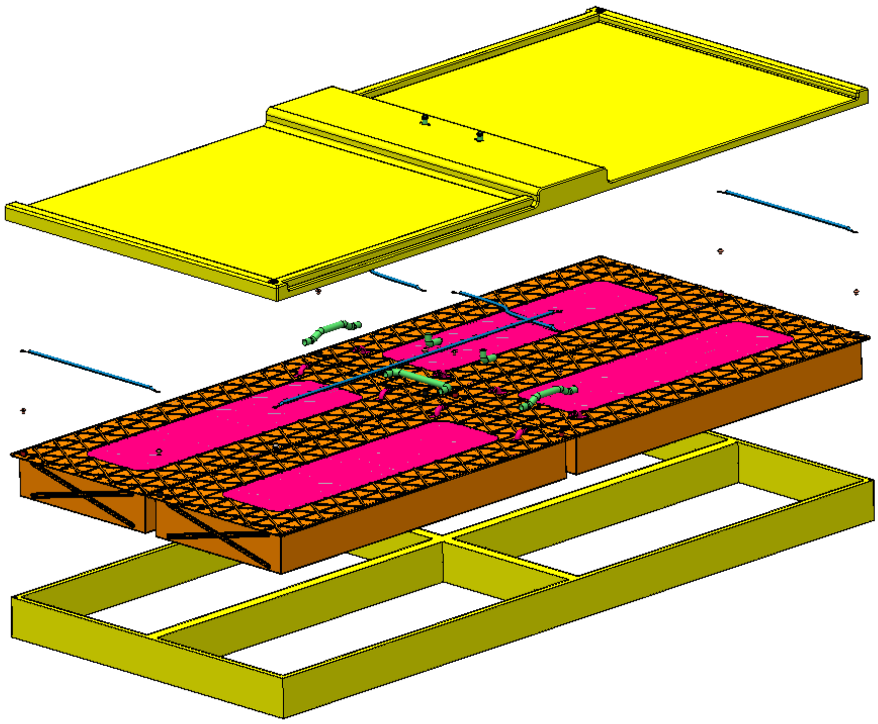


Figure 2.14: Exploded view of battery pack

2.4 Conclusion

The geometrical and energy parameters are reported in table 2.5.

Parameter	Module	Pack
Mass [kg]	117.1	490.5
Volume [l]	51.7 l	217.7 l
Length [mm]	1110	2224
Width [mm]	502	1008.4
Height [mm]	117	164.5
Total energy [Wh]	25077	100310

Table 2.5: Geometrical parameters

At this point is possible to compute the battery pack parameters to evaluate its performance (table 2.6)

Parameter	**Module**	**Pack**
Gravimetric energy density [Wh/kg]	214.2	204.5
Volumetric energy density [Wh/l]	485	460.8
Volumetric efficiency [-]	81.3%	77.2%

Table 2.6: Performance metrics

As the parameters demonstrate, both the battery and the module exhibit exceptional efficiency in terms of energy density and volumetric performance, ensuring optimal capabilities to meet all power and capacity requirements.

To enhance the results, the following approaches can be implemented:

- Conduct a FEM analysis to evaluate the pack's behavior, identifying its strengths and weaknesses. Based on the findings, explore the possibility of reducing the thickness of structural components or other materials to improve stiffness or decrease weight.

- Perform a thermal analysis to assess the efficiency of the cooling system and determine if its cooling capacity needs enhancement, or if it's feasible to reduce its size—especially the cold plate—thereby lowering the overall weight.

Report 3

FEM modeling and modal analysis of a cantilever beam

3.1 Introduction

The aim of the project is to understand and put in practice the fundamental concepts of the modal analysis. The effect of the damping coefficient and of the elastic modulus on the frequency response of a mechanical structure are studied, to analytically recreate the results obtained experimentally in the laboratory.

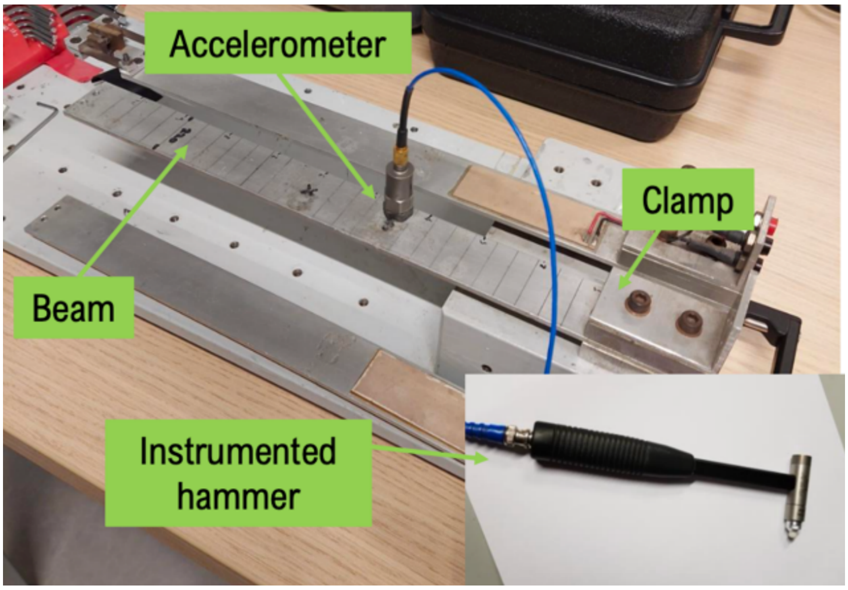


Figure 3.1: Test bench

The analyzed system is an aluminum beam, clamped on one side, which is free to vibrate on the opposite side. The instrumented hammer features a piezoelectric material at its tip, which converts the applied force into a voltage generation. When the material is unloaded, its fibers are randomly oriented. Upon the application of a force in a specific direction, the fibers align parallel to the direction of the force, with each fiber having a positive and a negative pole. As soon as the force is applied, the fibers align in the same direction, causing all the positive poles to gather on one side and the negative poles on the other. This behavior is reversible: the material generates an electrical signal when subjected to mechanical deformation, and it can also undergo deformation when an electric field is applied.

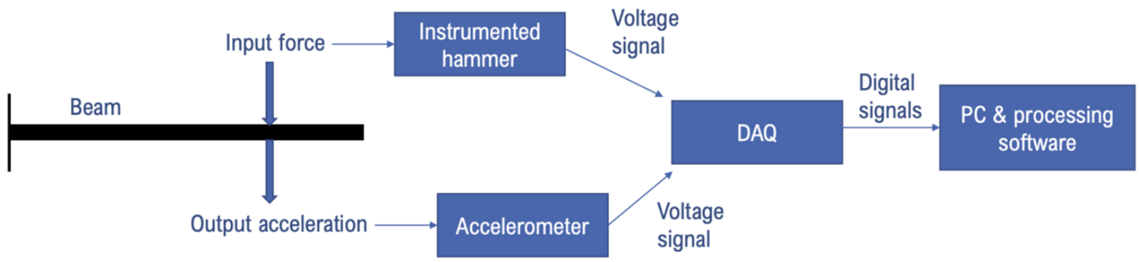


Figure 3.2: Experimental process

3.2 Method

3.2.1 Production of the FEM model of the beam

To discretize in finite elements, a set of parallel lines has been used on the test bench along the longitudinal axis of the structure. The beam (with estimated length of 280 mm) has been discretized in 28 elements along z-axis, with the accelerometer placed on the 11th parallel line as a direct consequence.

The following assumptions have been made:

- the material is homogeneous and isotropic;
- the bending on xz and yz are decoupled;
- the analysis is simplified by studying the equivalent $2D$ system.

A Matlab file contains the time history of the structure (experimentally produced in laboratory). It includes the frequency response function (transfer function), that has an excitation produced on the beam as input and the corresponding response in terms of acceleration as output.

The differential equation that describes the behavior of the system is the following one:

$$[M][\ddot{q}] + [C][\dot{q}] + [K][q] = [F] \tag{3.1}$$

After this procedure, stiffness matrix $[K]$ and mass matrix $[M]$ are known, as results of assembling 4x4 matrices coming from each single element in which the system has been discretized previously.

The solution of the system behavior can be expressed as the sum of a forced response and a free response. Since the free response is the solution of the homogeneous equations, it is the simplest one to be studied. On this purpose, it has been considered:

- undamped system $\implies$ $[C] = 0$ which, as simplification, implies no energy dissipation during the motion;
- homogeneous associated problem $\implies$ $[F] = 0$

The free response equation is expressed as

$$[M][\ddot{q}] + [K][q] = 0 \tag{3.2}$$

The solution of this equation is in the complex exponential form:

$$\begin{cases} \{q(t)\} = \{q_0\} \cdot e^{j\omega t} \\ \{\dot{q}(t)\} = j\,\omega\,\{q_0\} \cdot e^{j\omega t} \\ \{\ddot{q}(t)\} = -\omega^2\,\{q_0\} \cdot e^{j\omega t} \end{cases} \qquad \implies \qquad \big(-[M]\,\omega^2 + [K]\big) \cdot \{q_o\} = 0$$

Since $q_0 = 0$ is a trivial solution, the focus is on the subsequent equation:

$$[K] - \omega^2\,[M] = 0 \tag{3.3}$$

Assuming $\lambda = \omega^2$, the equation 3.3 can be rewritten as:

$$[K] - \lambda[M] = 0 \tag{3.4}$$

3.2.2 Solution of eigenproblem

From the equation 3.4, the eigenvectors and the eigenvalues can be extracted using Matlab function *eig*().

MATLAB script - Mode shapes

```
[eigvectors, eigvalues] = eig(Kc,Mc);

K_modal = eigvectors' * Kc * eigvectors;
M_modal = eigvectors' * Mc * eigvectors;

K_modal = K_modal .* eye(length(K_modal));
M_modal = M_modal .* eye(length(M_modal));
```

In the former Matlab code, Kc and Mc are intended as $[K]$ and $[M]$ which are 56X56 matrices. Each eigenvalue λ_i is grouped in a vector λ and they correspond to the square of natural frequencies of the structure. The eigenvectors are grouped in a square matrix $[\Phi]$, characterized by the following form:

$$\Phi = \begin{bmatrix} \Phi_1 & \Phi_2 & \dots & \Phi_{56} \\ \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

Eigenvectors describe how the structure is vibrating when it is working in correspondence of the natural frequency. If the previous matrix and his transposed are applied to the initial stiffness and mass matrices, the allow us to switch from space domain to the modal one:

$$\begin{cases} [k_{diag}] & = [\Phi]^T[K][\Phi] \\ [m_{diag}] & = [\Phi]^T[M][\Phi] \end{cases}$$

Both mass and stiffness matrix are diagonal: since terms outside of the main diagonal are negligible, a command *eye*() is applied on Matlab having, as final result, the removal of these terms.

The consequence of what has been described until now is the creation of a set of n second order differential equation that are all decoupled (and then independent) one from the other. This was not true for the canonical base because the physical displacements of each element of the beam are not decoupled. This set of independent differential equations can be solved as a system, making also the computation quite resource-consuming.

From the physical point of view this means that the system can be treated as a set of n masses (those contained in the mass matrix in modal form), each of them connected to the ground by means of spring with a correspondent stiffness value (those contained in the stiffness matrix in modal form).

3.2.3 Modal analysis for the definition of the modal damping matrix

So far, an equivalent undamped system has been studied by decoupling the equations through the modal transformation.

Considering now the damped system, we can write the dynamic equation

$$[M][\ddot{q}] + [C][\dot{q}] + [K][q] = [T_i][F] \tag{3.5}$$

Applying the modal transformation to the dynamic system we obtain:

$$[\Phi]^T[M][\Phi][\ddot{\zeta}] + [\Phi]^T[C][\Phi][\dot{\zeta}] + [\Phi]^T[K][\Phi][\zeta] = [\Phi]^T[T_i][F] \tag{3.6}$$

Where, as previous discussed, $[\Phi]^T[M][\Phi] = [m_{diag}]$ and $[\Phi]^T[K][\Phi] = [k_{diag}]$ are diagonal matrix, so we can describe the system as an equivalent discretized system where in each point act a group of mass, spring and dumper independent from other points. Supposing that the damping is small ad each point have equal damping factor ξ, $[c_{diag}]$ can be defined:

$$[c_{diag}] = \begin{bmatrix} c_{1,modal} & & \\ & \ddots & \\ & & c_{i,modal} \end{bmatrix}$$

Each element of the diagonal is obtained with the following equations:

$$c_{i,modal} = \xi \cdot c_{i,crit} \tag{3.7}$$

$$c_{i,crit} = 2 \cdot \sqrt{k_{i,modal} \cdot m_{i,modal}} \tag{3.8}$$

The damping factor ξ has to be tuned in order to obtain the peak magnitude of analytical results equal to experimental ones.

Afterwards, it is necessary to get back to the space domain using the modal expansion theorem to get the damping matrix $[C]$ of the dynamic system.

$$[C] = ([\Phi]^T)^{-1}[c_{diag}]([\Phi])^{-1} \tag{3.9}$$

Finally, all the variables of the dynamic equation (eq. 3.5) are defined.

MATLAB script - Modal analysis

```
%% MODAL ANALYSIS

C_crit_modal = 2.*sqrt(K_modal.*M_modal);

xi_const = 0.01;

xi_modal = xi_const .* eye(length(K_modal));

C_modal = xi_modal .* C_crit_modal;

Cc = inv(eigvectors')*C_modal*inv(eigvectors);
```

3.2.4 State Space representation

The state-space representation is used to describe the dynamic system. It is a mathematical framework which uses differential equations (expressed in matrix form with state), input and output variables. State variables represent the smallest set of information needed to describe the system's state at any given time and the number of state variables is equal to the system's order.

The *State-equation* describes the dynamic of the system:

$$\dot{x}(t) = Ax(t) + Bu \tag{3.10}$$

Where:

- $x(t)$: state vector (nx1)
- u: input vector (mx1)
- A: dynamic matrix (nx1)
- B: inputs matrix (nxm)

The *Output-equation* describes the output behavior:

$$\dot{y}(t) = Cx(t) + Du \tag{3.11}$$

Where:

- $y(t)$: outputs vector (qx1)
- C: output matrix (qxn)
- D: direct I/O matrix (qxp)

Considering the dynamic equation (eq. 3.5) we can write the *State-equation* as follow:

$$\begin{cases} \{\ddot{x}\} = -[M]^{-1}[C]\{\ddot{x}\} - [M]^{-1}[K]\{x\} + [M]^{-1}\{F\} \\ \{\ddot{x}\} = \{\ddot{x}\} \end{cases} \tag{3.12}$$

In matricial form:

$$\begin{Bmatrix} \ddot{x} \\ \dot{x} \end{Bmatrix} = \begin{bmatrix} -[M]^{-1}[C] & -[M]^{-1}[K] \\ I & 0 \end{bmatrix} \begin{Bmatrix} \dot{x} \\ x \end{Bmatrix} + \begin{bmatrix} -[M]^{-1} \\ 0 \end{bmatrix} \begin{Bmatrix} F \\ \end{Bmatrix} \tag{3.13}$$

So, the matrices A and B can be extrapolate:

$$A = \begin{bmatrix} -[M]^{-1}[C] & -[M]^{-1}[K] \\ I & 0 \end{bmatrix} \tag{3.14}$$

$$B = \begin{bmatrix} -[M]^{-1} \\ 0 \end{bmatrix} \tag{3.15}$$

Regarding the output equation (eq. 3.11), y vector are defined as follows. In the detail, the y is equal to the acceleration where the accelerometer is placed.

$$y = \ddot{x}_{acc} \tag{3.16}$$

At this point C and D are equal to the row of A and B, respectively, in correspondence of $\ddot{x}_{acc}$.

The implementation on Matlab is presented below.

MATLAB script - State space representation

```
%% State Space
M_inv = inv(Mc);
n_in = n_hammer;
```

```

A                  = [-M_inv*Cc   -M_inv*Kc; eye(length(Kc))
  ↪  zeros(length(Kc))];
B                  = [M_inv; zeros(length(Kc))];
B                  = B(:,2*(n_in-1)-1);
C                  = A((n_acc-1)*2-1,:);
D                  = B((n_acc-1)*2-1,:);

system = ss(A,B,C,D);
[mag , phase, W] = bode(system,{x(1).*(2*pi),x(end).*(2*pi)});
f = W./(2*pi)
```

3.2.5 Tuning of the model parameters trough a comparison with experimental results

The equivalent response function, obtained in Matlab, is going to be compared with experimental results (coming from the experimental modal analyzer) and tuned in a proper way, acting on stiffness and damping values, in order to get as close as possible with experimental results.

At this point, the experimental data and analytical results are plotted in the same graph in order to compare them.

To match the analytical model to experimental behavior same parameters are tuned:

- Elastic Modulus [E]: to align peaks on x-axis
- Damping coefficient [ξ]: to align peak on y-axis

The final results are reported below.

$$E = 65 \cdot 10^9 N/m^2 \qquad \xi = 0.01$$

MATLAB script - Results plotting

```
%% Comparison with Experimental data
% use the file Sq11_sn.mat to load experimental results

load("Sq11_sn.mat")

```

```
y = sqrt( real( FRF_Point2_Point1.y_values.values ).^2 + imag(
  ↪  FRF_Point2_Point1.y_values.values ).^2 );
x = FRF_Point2_Point1.x_values.start_value :
  ↪  FRF_Point2_Point1.x_values.increment :
  ↪  FRF_Point2_Point1.x_values.increment * (
  ↪  FRF_Point2_Point1.x_values.number_of_values - 1 );

% Plot
figure(2)
semilogx(x,20.*log10(y),LineWidth=lw)
hold on
semilogx(f,20.*log10(mag(:,:)),LineWidth=lw)
xline(omega(end),LineWidth=lw-1)
grid on
legend("Experimental","Analytical","Frequency range
  ↪  limit","FontSize",text_size, 'Location',"northwest" )
xlabel("Frequency [Hz]", "FontSize",text_size)
ylabel("Magnitude [dB]", "FontSize",text_size)
```

3.3 Results and discussion

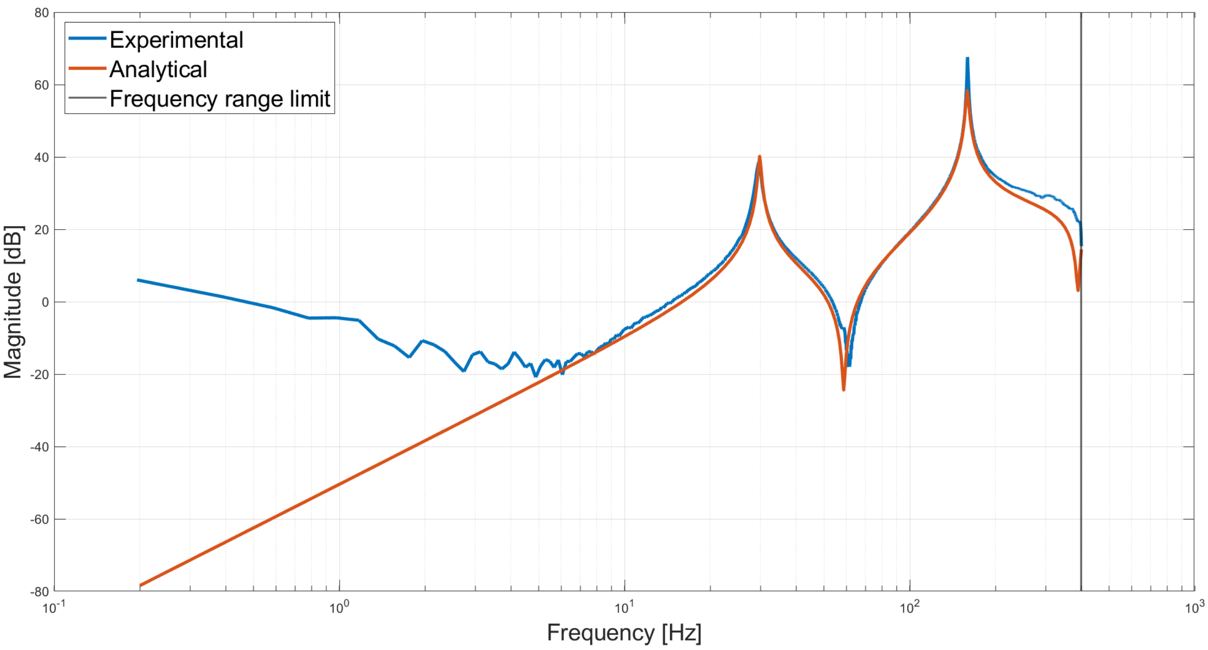


Figure 3.3: Bode plot

A large amount of energy is associated to the first modes of vibration, while the ones at higher frequencies are mainly compensated by inertia (only a small amount of energy is associated to them). This is the exact reason why the frequency range of interest for our case study goes from 0 to 400 Hz.

For frequencies close to 400 Hz, the numerical simulation approximates the experimental one with a little bit higher damping action. For lower frequencies, below 10 Hz, the numerical approximation behaves in a quite different way with respect to the experimental one: the first one increases with a slope of about $40dB/dec$ while experimental plot shows a decreasing slope of $-20dB/dec$. Considering that the two models behave almost identically in correspondence of the two natural frequencies (where our concerns are centered), the figure 3.3 demonstrates that the numerical trend has been properly tuned.

3.4 Conclusion

The modelization has been successful, in particular between the two natural frequencies. The discrepancy of the results before and after them are related to many factors, which are presented below.

- Not sufficiently accurate discretization: it could be not abele to capture accurately the material behavior, especially at lower frequencies.
- Experimental setup and instrumentation (eg. accelerometer) may lead to some errors, not considered in the simulation.
- The hypothesis of a constant damping factor in all the material could affect the analytical results, because in reality it could slightly vary.
- To get closer results, a higher number of nodes or a higher number of eigenvectors could be considered.

To improve the simulation, a higher number of nodes or eigenvalues, and consequently bigger mass and stiffness matrices, could be considered.